

MATH 2211, SPRING 2026: HOMEWORK 9

- (1) Suppose that $T : V \rightarrow V$ is an F -linear transformation with $\dim_F V = n$. Choose a basis $\mathcal{B}$ for V , and set $A = [T]_{\mathcal{B}}$. Show that the characteristic polynomial $\text{ch}_A(x)$ does not depend on which basis you chose.

Definition 1. If $\lambda \in F$ and $A \in M_{n \times n}(F)$, the λ -**eigenspace for A** $E_{\lambda}(A)$ is the kernel of $A - \lambda I_n$. The **geometric multiplicity of λ** (with respect to A) is the dimension of $E_{\lambda}(A)$.

- (2) Find the eigenvalues of

$$A = \begin{bmatrix} 7 & 4 & 4 \\ 0 & -1 & 0 \\ -8 & -4 & -5 \end{bmatrix} \in M_{3 \times 3}(\mathbb{C}).$$

Compute bases for its eigenspaces, and then diagonalize A or explain why this is not possible.

- (3) Find the eigenvalues of

$$B = \begin{bmatrix} -1 & 0 & -3 & 0 \\ 3 & 5 & 3 & -3 \\ 0 & 0 & 2 & 0 \\ 3 & 6 & 3 & -4 \end{bmatrix} \in M_{4 \times 4}(\mathbb{C}).$$

Compute bases for its eigenspaces, and then diagonalize B or explain why this is not possible.

Definition 2. If $\lambda \in F$ and $A \in M_{n \times n}(F)$, the **algebraic multiplicity of λ** (with respect to A) is the largest power of $x - \lambda$ that divides $\text{ch}_A(x)$.

- (4) Find a *non*-diagonalizable matrix $A \in M_3(\mathbb{R})$ such that, for all $\lambda \in \mathbb{R}$, the geometric multiplicity of λ equals the algebraic multiplicity of λ .
- (5) Fix $(n-1)$ -vectors $v_1, \dots, v_{n-1} \in F^n$ and $i \in \{1, \dots, n\}$. Consider the function

$$S_{(v_1, \dots, v_{n-1})}^i : F^n \rightarrow F$$

$$w \mapsto \det([v_1 \ v_2 \ \cdots \ v_{i-1} \ w \ v_i \ \cdots \ v_n])$$

In other words, we are fixing all the columns of a matrix except the i -th column, and viewing the determinant as a function of that column.

- (a) Show that $S_{(v_1, \dots, v_{n-1})}^i$ is an F -linear transformation, and conclude that there exist elements

$$C_{i,j}(v_1, \dots, v_{n-1}) \in F$$

for $j = 1, \dots, n$ such that

$$S_{(v_1, \dots, v_{n-1})}^i \left(\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \right) = C_{i,1}(v_1, \dots, v_{n-1})a_1 + C_{i,2}(v_1, \dots, v_{n-1})a_2 + \cdots + C_{i,n}(v_1, \dots, v_{n-1})a_n$$

- (b) By taking $w = e_j$, show that $C_{i,j}(v_1, \dots, v_{n-1})$ depends only the $(n-1) \times (n-1)$ -matrix obtained by taking the $n \times (n-1)$ matrix

$$A = [v_1 \ \cdots \ v_{n-1}]$$

and removing the j -th row: In other words, $C_{i,j}$ is a certain function on $(n-1) \times (n-1)$ -matrices.

(6) Consider the function $C_{i,j}$ from the previous problem, and think of it as a function on $(n-1) \times (n-1)$ -matrices.

(a) Show that $C_{i,j}$ satisfies all the properties of the $(n-1) \times (n-1)$ determinant *except* that $C_{i,j}(I_{n-1}) = (-1)^{i+j}$.

(b) Use this to conclude that, for any square matrix $A \in M_{n \times n}(F)$, we have

$$\det(A) = (-1)^{i+1} (a_{i,1}|A_{i,1}| - a_{i,2}|A_{i,2}| + \cdots + (-1)^{j-1} a_{i,j}|A_{i,j}| + \cdots + (-1)^{n-1} a_{i,n}|A_{i,n}|)$$

where $A_{i,j}$ is as in Lecture 26.

Definition 3. The **adjugate matrix** $\text{adj}(A)$ is the matrix whose (i, j) -th entry is $(-1)^{i+j}|A_{i,j}|$.

(7) Show that $A \cdot \text{adj}(A) = \det(A)I_n$. Use this to give a different proof that A is invertible if and only if $\det(A) \neq 0$, and give a formula for A^{-1} in terms of the adjugate matrix.

This expression for the inverse is called Cramer's rule.